BANKING CUSTOMER & RISK ANALYTICS

Portfolio Insights & ML Risk Modeling

Exploratory data analysis, operational recommendations, and an initial loan-default classification workflow



PROJECT PRESENTATION • EDA + MACHINE LEARNING

Executive overview

A cross-functional view of balances, branches, lending, payments, and early ML results.

690K+
Records across core datasets

500K
Transactions analyzed

₹6.02B
Transaction value reported

30K
Loan records modeled

Portfolio

Most positive balances sit below $50K; fixed deposits carry the highest average positive balance, while loan accounts have the largest negative average.

Operations

Branch account volumes are relatively uniform, but staffing varies substantially. Capacity alignment is worth investigating before reallocating staff.

Risk & payments

Loan default risk needs deeper segmentation. UPI dominates transaction volume, and payment success monitoring remains operationally important.

Note: figures and currencies are reproduced from the supplied EDA summary; validate currency conventions and metric definitions before external use.

1. Account balance & portfolio distribution

Balance distribution is concentrated at lower positive balances, with a wide tail of both negative and high positive exposure.

$0–$50K

Most accounts fall in this range

$10K–$30K

Peak account-volume band

~$125K

Fixed deposit avg. balance

~-$100K

Loan account avg. balance

What stands out

- Balances span approximately -$200K to +$250K.
- Savings average ~$38K; checking averages ~$18K.
- Credit-card balances average roughly -$7K.

Recommended next analysis

Separate deposit balances from credit exposure. Review outliers, product mix, and high-balance customer concentration before making wealth-management recommendations.

Currency label in the supplied summary is '$'; verify whether balances are actually USD or another currency before publishing.

2. Branch network & workforce efficiency

Footprint and staffing patterns point to opportunities for a capacity review—not an automatic headcount transfer.

26
Branches in South region

22
Branches in North region

12
Branches in Central region

13
Branches in Maharashtra

Operational signal

Branch account volumes are close to 3,370–3,449, while reported employee counts range from ~32 in Bhopal to 167 in Aurangabad.

Action to validate

Compare accounts and transactions per employee, service-level performance, opening hours, local demand, and branch roles before recommending staffing changes.

Avoid labeling a branch overstaffed from employee count alone; branch complexity and service workload may differ.

3. Lending & credit risk health

Loan volume is concentrated in active accounts and home loans; risk segmentation should go beyond income bands.

~21K
Active loans

~8.6K
Closed loans

~452
Loans labeled Defaulted

~10.6K
Home loans (largest product)

Income-tier finding

Default rates range from approximately 4.23% to 5.61%. The ₹10L–₹20L income band has the largest volume (15,289 loans) and highest reported default rate (5.61%).

Risk-management implication

Income alone may not separate risk well. Evaluate credit score, debt-to-income ratio, loan type, affordability, and repayment history. Validate definitions and target labels.

Data consistency check: the supplied summary reports ~452 defaulted loan statuses (~1.5% of 30K), while default_flag has 1,617 'Yes' records (5.39%). These are different labels/definitions and must not be conflated.

4. Transactions & payment channels

UPI dominates volume; credit transactions dominate monetary value. Reliability should be monitored by channel and value.

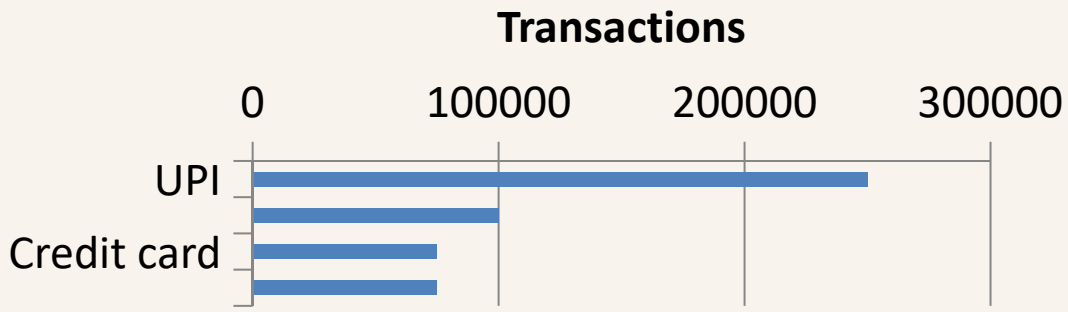
500K
Total transactions

₹6.02B
Reported transaction value

~250K
UPI transactions

~460K
Successful transactions

Approx. volume by payment channel



Value and reliability

Credit transactions contribute ~83% of value (avg. ₹50,490.01); debit transactions contribute ~80% of volume (avg. ₹2,505.35). Channel failure rates are reported at roughly 4.8%–5.0%.

Action: track failure rate by channel, transaction value, time, and gateway; prioritize monitoring for high-volume UPI and high-value credit flows.

Key actionable recommendations

Convert the descriptive findings into testable operational actions.

01 Workforce capacity review

Compare accounts and transactions per employee, wait times, SLA, and branch workload before adjusting staffing.

02 Wealth and deposit retention

Segment high-balance and fixed-deposit customers; evaluate suitable cross-sell and retention opportunities.

03 Better credit-risk segmentation

Assess credit score, affordability, loan type, debt-to-income, and repayment history; test model stability across groups.

04 Payment-channel reliability

Monitor channel-level success, failure reasons, transaction value, and time-to-resolution; prioritize UPI volume and high-value flows.

Prioritize recommendations only after validating source definitions, operational constraints, and customer outcomes.

Machine learning: loan-default classification

A baseline classification workflow built from 30,000 loan records joined with selected customer attributes.

30,000
Loan records

5.39%
Default_flag = Yes

80 / 20
Stratified train/test split

10
Selected input features

Target and safeguards

Target: default_flag (No/Yes). Excluded IDs, loan_status, and remaining_balance from the initial features to reduce leakage risk. Numeric inputs were scaled; categorical inputs one-hot encoded.

Models trained

Logistic Regression with class_weight='balanced' and Random Forest with class_weight='balanced'. Test-set metrics include precision, recall, F1-score, accuracy, and ROC-AUC.

Note: the observed target distribution is imbalanced (1,617 defaults vs 28,383 non-defaults); accuracy alone is not a sufficient success measure.

ML results: accuracy alone would mislead

Held-out test set: 6,000 loans, including 323 positive default labels.

Metric	Logistic Regression	Random Forest
Accuracy	70.23%	93.57%
Precision	12.12%	20.56%
Recall	72.45%	6.81%
F1-score	20.76%	10.23%
ROC-AUC	77.77%	76.48%

Recall matters

Logistic Regression identified 234 of 323 actual defaults at threshold 0.50; Random Forest identified only 22.

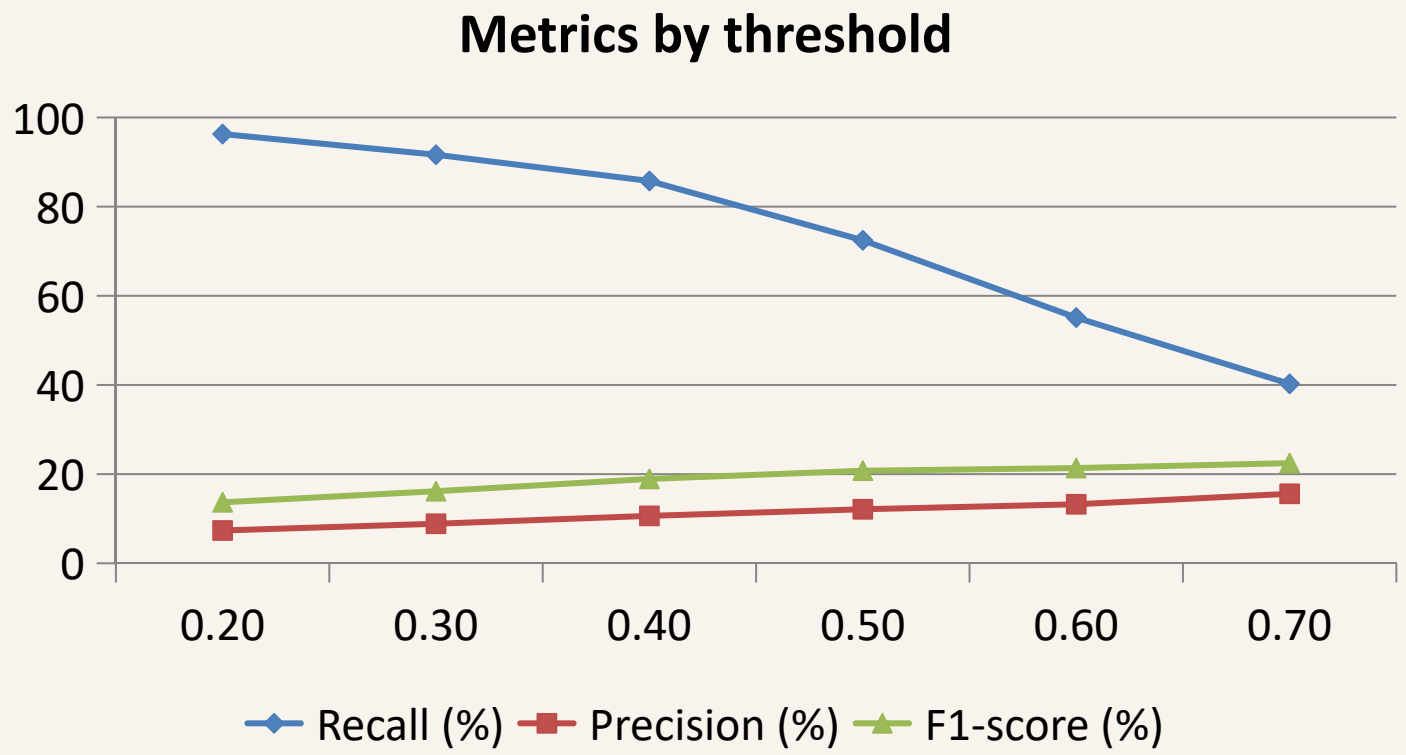
Interpretation

Random Forest's higher accuracy largely reflects predicting the majority non-default class. Logistic Regression is the stronger baseline for catching defaults at threshold 0.50.

Caution: these are preliminary results from one train/test split. Validate using cross-validation, threshold selection on validation data, calibration, fairness analysis, and realistic cost assumptions before drawing operational conclusions.

Threshold tuning: a business trade-off

Lower thresholds catch more defaults but increase false alarms. Threshold choice should reflect review capacity and the cost of missed defaults.



Threshold 0.20

Recall 96.28%; 12 defaults missed, but 3,914 false alarms.

Threshold 0.70

F1-score is highest among tested thresholds (22.47%), but 193 defaults are missed.

The 0.50 threshold is retained as the baseline for reporting; threshold 0.70 has the best F1 among this tested grid, not necessarily the best business outcome.

Feature insights from both models

Credit score ranks first in both models, but feature importance is descriptive and does not establish causality.

Logistic Regression coefficients

Largest absolute coefficient: credit_score (-1.2782). Other top coefficients include occupation, marital-status, and customer-segment indicators. Coefficient direction is relative to encoded reference categories.

Random Forest importance

Top features: credit_score (0.3177), interest_rate (0.1544), EMI (0.1078), annual_income (0.0984), age (0.0874), and loan_amount (0.0835).

What this suggests

Creditworthiness and loan terms merit closer investigation in the next iteration.

What it does not prove

Importance does not mean causation, and these findings do not establish suitability for real lending decisions.

Recommended next: permutation importance on held-out data, calibration, error analysis, and subgroup fairness checks.

Limitations & next steps

Turn the initial analysis into a reproducible, carefully validated portfolio-risk project.

- Validate currency units and source metric definitions.
- Reconcile loan_status='Defaulted' with default_flag='Yes' before reporting a single default rate.
- Keep threshold selection on a validation set, not the final test set.
- Use cross-validation and tune class weights / model hyperparameters.
- Assess precision-recall curve and probability calibration.
- Check leakage, data drift, and data quality over time.
- Evaluate subgroup error rates and fairness before considering deployment.
- Translate model scores into a review workflow—not automatic lending decisions.

Conclusion: the project combines portfolio EDA, operational recommendations, and a first imbalanced-classification workflow; careful validation is the next milestone.